

Seventh Semester					
Group	Paper Code	Paper	L	P	Credits
Theory Papers					
HS/MS	MS-401	Principles of Entrepreneurship Mindset	2		2
PCE		Programme Core Elective Paper (PCE – 4)			4
PCE		Programme Core Elective Paper (PCE – 5)			4
EAE / OAE		Emerging Area / Open Area Elective Paper (EAE – 3 / OAE – 3)			4
EAE / OAE		Emerging Area / Open Area Elective Paper (EAE – 4 / OAE – 4)			4
EAE / OAE		Emerging Area / Open Area Elective Paper (EAE – 5 / OAE – 5)			4
Practical / Viva Voce					
PC / Project	ES-451	Minor Project**			3
PC / Internship	ES-453	Summer Training Report - 2 *			1
Total					26

***NUES:** Comprehensive evaluation of the Summer Training Report – 2 (after 6th Semester) shall be done by the committee of teachers, constituted by the Academic Programme Committee, out of 100. The training shall be of 4 to 6 weeks duration. The training can be under the mentorship of a teacher of the institute.

******The student shall be allocated a supervisor / guide for project work at the end 6th semester by the department / institution, the project shall continue into the 8th semester. In the 7th semester evaluation, the criteria for evaluation shall be conceptualization of the project work, the background study / literature survey and identification of objectives and methodology to be followed for project. 40 marks evaluation for the Teachers' Continuous Evaluation / Internal Assessment shall be done by concerned supervisor while the term end examination of 60 marks shall be conducted by the supervisor concerned and the external examiner deputed by the Examinations Division. In the absence of the supervisor, the Director of the Institution / Head of the Department can assign the responsibility of the supervisor (for purpose of examinations) to any faculty of the Institution / Department.

Eight Semester					
Group	Paper Code	Paper	L	P	Credits
Practical / Viva Voce[%]					
PC / Project	ES-452	Major Project – Dissertation and Viva Voce [#]			18
	ES-454	Project Progress Evaluation*			2
PC / Internship	ES-456	Internship Report and Viva Voce [#]			18
	ES-458	Internship Progress Evaluation*			2
Total			0	0	20

***NUES:** Comprehensive evaluation by the committee of teachers, constituted by the Academic Programme Committee, out of 100.

%By default every student shall do the project work (ES-452 and ES-454). A student shall either be allowed to do a project work (ES-452 and ES-454) or an internship (ES-456 and ES-458). The student must apply for approval to do internship before the commencement of the 8th semester to the institute, and only after approval of Principal / Director of the institute through Training and Placement Officer of the institute, shall proceed for internship.

#Students may be allowed to do internship in this semester in lieu of Major project. The students allowed to proceed for internship shall be required to maintain a log-book of activities performed during internship. The same has to be countersigned by the mentor at the organization where internship is completed.

Intellectual Property Rights			
		L	P
		4	4

Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
CSE/IT/CST/ITE	7	PCE	PCE-5	CIE-419

Marking Scheme:												
1. Teachers Continuous Evaluation: 25 marks												
2. Term end Theory Examinations: 75 marks												
Instructions for paper setter:												
1. There should be 9 questions in the term end examinations question paper.												
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.												
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.												
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.												
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.												
Course Objectives :												
1.	To Understand, define and differentiate different types of intellectual properties (IPs) and their roles in contributing to organizational competitiveness and Understanding the Framework of Strategic Management of Intellectual Property (IP).											
2.	To Appreciate and appraise different IP management (IPM) approaches and describing how pioneering firms initiate, implement and manage IPM programs.											
3.	To Explain how to derive value from IP and leverage its value in new product and service development.											
4.	To Expose to the Legal management of IP and understanding of real life practice of IPM.											
Course Outcomes (CO)												
CO 1	To Identify different types of Intellectual Properties (IPs), the right of ownership, scope of protection as well as the ways to create and to extract value from IP and Recognize the crucial role of IP in organizations of different industrial sectors for the purposes of product and technology development.											
CO 2	To Identify activities and constitute IP infringements and the remedies available to the IP owner and describe the precautions steps to be taken to prevent infringement of proprietary rights in products and technology development and Be familiar with the processes of Intellectual Property Management (IPM) and various approaches for IPM and conducting IP and IPM auditing and explain how IP can be managed as a strategic resource and suggest IPM strategy.											
CO 3	To be able to anticipate and subject to critical analysis arguments relating to the development and reform of Geographical Indications of goods and their likely impact on creativity and innovation.											
CO 4	To be able to demonstrate a capacity to identify, apply and assess ownership rights and marketing protection under intellectual property law as applicable to information, ideas, new products and product marketing											
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)												
	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO 1	2	1	-	2	-	1	1	1	-	1	-	-
CO 2	2	1	-	2	-	1	1	-	-	1	-	-
CO 3	2	1	-	-	-	1	1	-	-	1	-	-
CO 4	2	1	-	-	-	1	1	1	-	1	-	-
UNIT-I												
Understanding & Overview of IPR Regime & Patents: Introduction, Need of Intellectual Property Rights, Types of Intellectual property, Rationale for protection of IPR, International Organizations, Agencies and Treaties,												

Importance of Intellectual Property Rights, IPR in India: Genesis and development, Impact of IPR on development, health agriculture and genetic resources in India.

Patents: Definition of TRIPS, Kind of inventions protected by Patent, Process and Product Patent, Double Patent, Legal requirements for patents, Patent Application process, Searching a patent, Drafting and Filing of a Patent, Type of patent applications, Patent Specification and Claim, Management of IP Assets and IP Portfolio, Layers of International Patent System: National, Regional and International options

UNIT-II

Trademarks & Copyrights: Rights of Trademark, Purpose and Function of Trademark, Trademark registration, Trademark Acquisition, Trademark protection, Trademark Registration process.

Law of Copy Rights, Rights and Protection covered by Copyright, Infringement of Copyright under Copyright Act, Distinction between related rights and copyright, Celebrity rights, Academic integrity or Plagiarism

UNIT-III

Geographical Indications of Goods & Traditional Knowledge: Types of Geographical Indications of Goods, Why Geographical Indications of Goods needs protection, How Geographical Indications of Goods gets protection, GI Laws, GI Act in India, Case Study: Three Recent famous registered GIs of India.

Examples of medicinal, bioprospecting and Indigenous knowledge Need for their Protection, Positive protection, Defensive protection and legal aspects.

UNIT – IV

Recent Developments & Institutional Capacity: New Development in Trademark Law, Copyright Law, Patent Law, Intellectual Property Audits, Protection of Computer Programs, Protection of Communication Technologies, Protection of Biotechnology, Reprography and Intellectual Property.

International Trademark law, international copyright law, International Patent law, International Development in Trade Secret Law.

IP Policy making and legislation, Examination and Registration Systems, International Standard Setting: WIPO & WTO.

Textbook(s):

1. Fundamentals of IP for Engineers, K. Bansal & P. Bansal, BS/BSP Books Publications, 2013
2. Intellectual Property Rights, Deborah. E. Bouchoux, 5th Edition, Delmar Cengage Learning, 2017.

References:

1. Intellectual Property Rights– Unleashmy The Knowledge Economy, Prabuddha Ganguli, Tate Mc-Graw Hill Publishing Company Ltd, 2001.

Mobile App Development			
		L	P
		3	3

Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
EAE	7	FSD-EAE	FSD-EAE-4	FSD-437T

Marking Scheme:

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

Instructions for paper setter:

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

Course Objectives :

- | | |
|----|--|
| 1. | To introduce students the fundamentals of mobile app development and its significance in the digital era. |
| 2. | To familiarize students with the Android and iOS platforms as key mobile app development platforms. |
| 3. | To provide students with hands-on experience in designing, developing, testing, and deploying mobile applications. |
| 4. | To expose students to advanced subjects in mobile app development, such as cross-platform development and upcoming trends. |

Course Outcomes (CO)

- | | |
|-------------|---|
| CO 1 | Understand the importance, principles of mobile app development, Identify and explain the characteristics and features of the Android and iOS platforms. |
| CO 2 | Develop mobile applications using programming languages relevant to the platforms and design user-friendly and visually appealing mobile app interfaces. |
| CO 3 | Implement data storage, synchronization, and location-based services in mobile apps. Test and debug mobile applications for optimal performance and functionality. |
| CO 4 | Demonstrate knowledge and understanding of cross-platform app development frameworks and Stay updated with emerging trends and future directions in mobile app development. |

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO 1	3	2	1	1	1	2	-	3	-	1	2	3
CO 2	3	2	1	1	1	2	-	3	-	1	2	3
CO 3	3	2	1	1	1	2	-	3	-	1	2	3
CO 4	3	2	1	1	1	2	-	3	-	1	2	3

UNIT I

Introduction to Mobile App Development: Overview, history, and importance, Mobile platforms and operating systems: Android and iOS, Mobile app development tools and environments, Introduction to programming languages for mobile app development

User Interface Design and Development: User interface (UI) design principles for mobile apps, UI components and layouts, designing for multiple screen sizes and resolutions, Implementing navigation and user interaction

UNIT II

Mobile App Development Technologies: Mobile app architecture and components, Backend integration and API consumption, Data storage and synchronization, Location-based services and mapping.

UNIT III

Mobile App Testing and Deployment: Testing methodologies for mobile apps, Debugging and error handling, App store submission and deployment process, App performance optimization and analytics

UNIT IV

Advances Topics in Mobile App Development: Mobile app security and privacy considerations, InCross-platform app development frameworks (e.g., React Native, Flutter), Emerging trends and future directions in mobile app development.

Textbooks:

1. "Android Programming: The Big Nerd Ranch Guide" by Bill Phillips and Chris Stewart
2. "iOS Programming: The Big Nerd Ranch Guide" by Christian Keur and Aaron Hillegass
3. "Head First Android Development: A Brain-Friendly Guide" by Dawn Griffiths and David Griffiths
4. "Beginning iOS Cloud and Database Development: Build Data-Driven Cloud Apps for iOS" by Thomas P. Fitzpatrick

Reference Books:

1. "Learning Swift: Building Apps for macOS, iOS, and Beyond" by Jonathon Manning, Paris Buttfield-Addison, and Tim Nugent
2. "Beginning Android Programming with Kotlin" by Jerome DiMarzio
3. "Designing Interfaces: Patterns for Effective Interaction Design" by Jenifer Tidwell
4. "iOS Human Interface Guidelines: Creating a Great User Experience" by Apple Inc.
5. "Android Studio Development Essentials: Android 10 Edition" by Neil Smyth
6. "Test-Driven Development with Python" by Harry Percival
7. "Mobile App Development with React Native: Build iOS and Android Apps with JavaScript" by Johnathan Horton and Muhammed Murtaza
8. "Mobile Design and Development: Practical concepts and techniques for creating mobile sites and web apps" by Brian Fling

Mobile App Development Lab	L	P	C
		2	1

Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
EAE	7	FSD-EAE	FSD-EAE-4	FSD-437P

Marking Scheme: 1. Teachers Continuous Evaluation: 40 marks 2. Term end Theory Examinations: 60 marks
Instructions: 1. The course objectives and course outcomes are identical to that of (Mobile App Development) as this is the practical component of the corresponding theory paper. 2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

Experiment 1: Setting up the Development Environment

- Install Android Studio and set up the Android development environment.
- Install Xcode and set up the iOS development environment.
- Create a basic "Hello World" app for Android and iOS platforms.

Experiment 2: User Interface Design and Development

- Design a simple user interface using XML for an Android app.
- Design a user interface using Interface Builder for an iOS app.
- Implement navigation between multiple screens in both Android and iOS apps.

Experiment 3: Database Integration in Mobile Apps

- Create a SQLite database and perform basic CRUD operations in an Android app.
- Implement Core Data framework for data storage in an iOS app.

Experiment 4: Web Services and API Consumption

- Retrieve data from a RESTful API and display it in an Android app.
- Consume a web service and display data in an iOS app using URLSession.

Experiment 5: Location-Based Services and Mapping

- Implement location tracking and display user location on a map in an Android app using Google Maps API.
- Integrate MapKit framework in an iOS app to show user location and add annotations on the map.

Experiment 6: Cross-Platform App Development with React Native or Flutter

- Develop a simple mobile app using React Native, utilizing components and navigation.
- Create a mobile app using Flutter, implementing UI elements and handling user input.

Experiment 7: Testing and Debugging

- Perform unit testing on key functionalities of an Android app using JUnit and Android Testing frameworks.
- Debug and test an iOS app using Xcode debugger and XCTest framework.

Experiment 8: App Deployment and Performance Optimization

- Package and deploy an Android app to Google Play Store.
- Package and deploy an iOS app to the App Store.
- Optimize app performance by analyzing and improving resource usage, network efficiency, and responsiveness.

Experiment 9: Augmented Reality (AR) Integration

- Integrate AR features into an Android app using ARCore or ARKit for iOS.

Experiment 10: IoT Integration in Mobile Apps

- Connect a mobile app with IoT devices, such as smart home devices, sensors, or wearables.

PHP Programming and MySQL	L	P	C
	3		3

Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
CSE-in-EA	7	OAE-CSE-EA	OAE-2	FSD-435T
EAE	7	FSD-EAE	FSD-EAE-3	FSD-435T

Marking Scheme:

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

Instructions for paper setter:

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

Course Objectives :

1. Learn how to take a static website and turn it into a dynamic website run from a database using PHP and MySQL.
2. Analyze the basic structure of a PHP web application and be able to install and maintain the web server, compile, and run a simple web application.
3. Learn how databases work and how to design one, as well as how to use php MyAdmin to work with MySQL.
4. Learn different ways of connecting to MySQL through PHP, and how to create tables, enter data, select data, change data, and delete data. Connect to SQL Server and other data sources.

Course Outcomes (CO)

- CO 1** Interpret the server side scripting PHP and create dynamic web pages.
- CO 2** Outline the advanced concepts of PHP and design web pages to authenticate users.
- CO 3** Develop server side programs using PHP and accessing database through PHP.
- CO 4** Design web pages to authenticate users using Cookies.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO 1	3	2	3	2	3	1	-	-	3	2	3	2
CO 2	3	2	3	2	3	1	-	-	3	2	3	2
CO 3	3	2	3	2	3	1	-	-	3	2	3	2
CO 4	3	2	3	2	3	1	-	-	3	2	3	2

UNIT-I

Introduction to PHP: Evaluation of PHP, Basic Syntax, Defining variable and constant, PHP Data type, Operator and Expression, Decisions and loop: Making Decisions, Doing Repetitive task with looping, Mixing Decisions and looping with Html, Function: What is a function, Define a function, Call by value and Call by reference, Recursive function, String Creating and accessing, String Searching & Replacing String, Formatting String, String Related Library function

UNIT-II

Array Anatomy of an Array, Creating index based and Associative array Accessing array, Element Looping with

Index based array, Looping with associative array using each () and foreach(), Some useful Library function, Handling Html Form with PHP Capturing Form, Data Dealing with Multi-value filed, and Generating File uploaded form, redirecting a form after submission

UNIT-III

Working with file and Directories: Understanding file& directory, Opening and closing, a file, Coping, renaming and deleting a file, working with directories, Creating and deleting folder, File Uploading & Downloading, Session and Cookie: Introduction to Session Control, Session Functionality What is a Cookie, Setting Cookies with PHP. Using Cookies with Sessions, Deleting Cookies, Registering Session variables, Destroying the variables and Session.

UNIT - IV

Introduction to RDBMS: Connection with MySql Database, Performing basic database operation (DML) (Insert, Delete, Update, Select), Setting query parameter, Executing queryJoin (Cross joins, Inner joins, Outer Joins, Self joins.)

Textbook(s):

1. Dave W Mercer, Allan Kent, Steven D Nowicki, David Mercer, Dan Squier, Wankyu Choi, "Beginning PHP".
2. RasmusLerdorf and Kevin Tatore , "Programming PHP"

References:

1. Learning PHP, MySQL, books by ' O' riley Press
2. PHP, MySQL and Apache by Julie C Meloni. Pearson Education

PHP Programming and MySQL Lab			
		L	P
		2	1

Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
CSE-in-EA	7	OAE-CSE-EA	OAE-2	FSD-435P
EAE	7	FSD-EAE	FSD-EAE-3	FSD-435P

Marking Scheme:

1. Teachers Continuous Evaluation: 40 marks
2. Term end Theory Examinations: 60 marks

Instructions:

1. The course objectives and course outcomes are identical to that of (PHP Programming and MySQL) as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

1. a. Install and configure PHP, web server, MYSQL
b. Write a program to print "Welcome to PHP".
c. Write a simple PHP program using expressions and operators.
2. Write a PHP program to demonstrate the use of Decision making control structures using:
a. If statement
b. If-else statement
c. Switch statement
3. Write a PHP program to demonstrate the use of Looping structures using:
a. while statement
b. do-while statement
c. for statement
d. foreach statement
4. Write a PHP program for creating and manipulating-
a. Indexed array
b. Associative array
c. Multidimensional array
5. a. Write a PHP program to-
i. Calculate length of string.
ii. Count the number of words in string without using string functions.
b. Write a simple PHP program to demonstrate use of various built-in string functions.
6. Write a simple PHP program to demonstrate use of Simple function and Parametrized function.
7. Develop web page with data validation.
8. Write simple PHP program to-
a. Set cookies and read it.
b. Demonstrate session management
9. Develop a simple application to-
a. Enter data into database.
b. Retrieve and present data from database.
10. Develop a simple application to Update, Delete table data from database.

Principles of Entrepreneurship Mindset			
		L	P
		2	2

Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
All	7	HS/MS	MS	MS-401

Marking Scheme:												
1. Teachers Continuous Evaluation: 25 marks												
2. Term end Theory Examinations: 75 marks												
Instructions for paper setter:												
1. There should be 9 questions in the term end examinations question paper.												
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.												
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.												
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.												
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.												
Course Objectives :												
1.	To understand basic aspects of establishing a business in a competitive environment											
2.	To apply the basic understanding to examine the existing business ventures											
3.	To examine various business considerations such as marketing, financial and teaming etc.											
4.	To assess strategies for planning a business venture											
Course Outcomes (CO)												
CO 1	Understand basic aspects of establishing a business in a competitive environment											
CO 2	Apply the basic understanding to examine the existing business ventures											
CO 3	Examine various business considerations such as marketing, financial and teaming etc.											
CO 4	Assessing strategies for planning a business venture											
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)												
	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO 1	2	2	1	2	-	2	-	-	1	2	3	2
CO 2	2	2	1	2	-	2	-	-	1	2	3	2
CO 3	2	2	1	2	-	2	-	-	1	2	3	2
CO 4	2	2	1	2	-	2	-	-	1	2	3	2
UNIT-I												
Entrepreneurial perspective: Foundation, Nature and development of entrepreneurship, importance of entrepreneurs, Entrepreneurial Mind, Individual entrepreneur Types of entrepreneurs, Entrepreneurship in India												
UNIT-II												
Beginning Considerations: Creativity and developing business ideas; Creating and starting the venture; Building a competitive advantage; Opportunity recognition, Opportunity assessment; Legal issues												
UNIT-III												
Developing Financial Plans: Sources of Funds, Managing Cash Flow, Creating a successful Financial Plan Developing a business plan												

UNIT - IV

Developing Marketing Plans: Developing a powerful Marketing Plan, E-commerce, Integrated Marketing Communications

Leading Considerations: Developing Team, Inviting candidates to join team, Leadership model

Textbook(s):

1. Robert D Hisrich, Michael P Peters & Dean A Shepherd, "Entrepreneurship" 10th Edition, McGraw Hill Education, 2018

References:

1. Norman M. Scarborough and Jeffery R. cornwell, "Essentials of entrepreneurship and small business management" 8th Edition, Pearson, 2016
2. Rajiv Roy, "Entrepreneurship", 2nd Edition, Oxford University Press, 2011
3. Sangeeta Sharma, "Entrepreneurship Development", 1st Edition, Prentice-Hall India, 2016
4. John Mullins, "The New Business Road Test: What entrepreneurs and investors should do before launching a lean start-up" 5th Edition, Pearson Education, 2017
5. Charantimath, Entrepreneurship Development and Small Business Enterprise, Pearson Education.

Web and Mobile Application Testing and Deployment			
L	P	C	
3			3

Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
EAE	7	FSD-EAE	FSD-EAE-5	FSD-439T

Marking Scheme:

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

Instructions for paper setter:

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

Course Objectives :

- | | |
|----|--|
| 1. | To introduce students to the concepts, methodologies, and best practices of web and mobile application testing and deployment. |
| 2. | To familiarize students with performance testing, optimization, and security considerations in web and mobile application development. |
| 3. | To equip students with the knowledge and skills to implement continuous integration, delivery, and deployment processes. |
| 4. | To enhance students' ability to effectively test, deploy, and maintain web and mobile applications in various environments. |

Course Outcomes (CO)

- | | |
|-------------|--|
| CO 1 | Demonstrate a clear understanding of the importance of testing and deployment in the software development life cycle and apply functional testing techniques to ensure the quality and reliability of web and mobile applications. |
| CO 2 | Perform performance testing, analyze results, and optimize web and mobile applications for better performance and Conduct security testing to identify vulnerabilities and implement necessary countermeasures. |
| CO 3 | Implement continuous integration, delivery, and deployment pipelines for efficient software development and deployment and apply industry best practices for deploying web and mobile applications in different environments. |
| CO 4 | Develop the ability to use appropriate tools and technologies for web and mobile application testing and deployment and collaborate effectively in teams to test, deploy, and maintain web and mobile applications. |

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO 1	3	3	3	2	2	3	-	3	3	2	2	3
CO 2	3	3	3	2	2	3	-	3	3	2	2	3
CO 3	3	3	3	2	2	3	-	3	3	2	2	3
CO 4	3	3	3	2	2	3	-	3	3	2	2	3

UNIT I

Introduction to Web and Mobile Application Testing and Deployment: Overview of web and mobile application testing and deployment, Importance of testing and deployment in software development life cycle,

Testing methodologies and strategies, Deployment models and techniques

Functional Testing of Web and Mobile Applications: Introduction to functional testing, Test case design techniques, Test automation frameworks and tools, Cross-browser and cross-platform testing, Mobile application testing

UNIT II

Performance Testing and Optimization: Performance testing concepts and objectives, Load testing and stress testing, Performance measurement and profiling tools, Performance optimization techniques, Mobile performance testing considerations.

UNIT III

Security Testing and Deployment Best Practices: Introduction to security testing, Security vulnerabilities and threats, Security testing techniques and tools, Secure deployment best practices, Compliance and regulatory considerations.

UNIT IV

Continuous Integration, Delivery, and Deployment: Introduction to continuous integration, delivery, and deployment, Continuous integration and build automation tools, Continuous delivery pipelines
Deployment strategies and techniques, Monitoring and error tracking in production

Textbooks:

1. "Software Testing: Principles and Practices" by Srinivasan Desikan and Gopalaswamy Ramesh
2. "Agile Testing: A Practical Guide for Testers and Agile Teams" by Lisa Crispin and Janet Gregory
3. "Software Testing": Yogesh Singh Cambridge University Press

Reference Books:

1. "Effective Software Testing: 50 Specific Ways to Improve Your Testing" by Elfriede Dustin, et al.
2. "Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation" by Jez Humble and David Farley
3. "Selenium WebDriver Recipes in Python: The problem-solving guide to Selenium WebDriver in Python" by Zed A. Shaw
4. "Hands-On Mobile App Testing: A Guide for Mobile Testers and Anyone Involved in the Mobile App Business" by Daniel Knott
5. "The Art of Application Performance Testing: Help for Programmers and Quality Assurance" by Ian Molyneux
6. "Web Performance Tuning: Speeding Up the Web" by Patrick Killelea
7. "The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws" by Dafydd Stuttard and Marcus Pinto
8. "OWASP Testing Guide" by The Open Web Application Security Project (OWASP)
9. "How to Break Software: A Practical Guide to Testing" by James A. Whittaker
10. "Performance Testing Guidance for Web Applications" by Microsoft Corporation
11. "Software Security: Building Security In" by Gary McGraw
12. "Continuous Integration: Improving Software Quality and Reducing Risk" by Paul M. Duvall

Web and Mobile Application Testing and Deployment Lab			
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Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
EAE	7	FSD-EAE	FSD-EAE-5	FSD-43P

Marking Scheme: 1. Teachers Continuous Evaluation: 40 marks 2. Term end Theory Examinations: 60 marks
Instructions: 1. The course objectives and course outcomes are identical to that of (Web and Mobile Application Testing and Deployment) as this is the practical component of the corresponding theory paper. 2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

1: Introduction to Web Technologies

Objective: Familiarize students with basic web technologies, HTML, CSS, and JavaScript.

Activities: Create a simple web page using HTML, style it with CSS, and add interactivity using JavaScript.

2: Web Application Testing

Objective: Introduce students to various testing techniques and tools for web applications.

Activities: Test a web application for functional and non-functional requirements; perform unit testing, integration testing, and regression testing.

3: Mobile Application Development

Objective: Enable students to develop mobile applications using a cross-platform framework.

Activities: Build a simple mobile application using a framework like React Native or Flutter.

4: Mobile Application Testing

Objective: Introduce students to the challenges and techniques of testing mobile applications.

Activities: Test a mobile application for usability, performance, compatibility, and security.

5: Web Application Deployment

Objective: Teach students the process of deploying a web application to a hosting environment.

Activities: Deploy a web application on a cloud hosting platform like Amazon Web Services (AWS) or Heroku.

6: Continuous Integration and Deployment

Objective: Introduce students to the concepts of continuous integration and deployment for web and mobile applications.

Activities: Set up a CI/CD pipeline using tools like Jenkins, Travis CI, or GitLab CI/CD.

7: Performance Testing for Web Applications

Objective: Teach students how to analyze and improve the performance of web applications.

Activities: Perform load testing, stress testing, and optimize the performance of a web application.

8: Security Testing for Web and Mobile Applications

Objective: Introduce students to the fundamentals of security testing for web and mobile applications.

Activities: Conduct vulnerability assessments, penetration testing, and analyze security issues in a web or mobile application.

Wireless Communication and Networks	L	P	C
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Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
CSE-NET	6	PC	PC	WMC-340T
EAE	6	WMC-EAE	WMC-EAE-1C	WMC-336T
IT	7	PCE	PCE-4	CIE-415T

Marking Scheme:

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

Instructions for paper setter:

1. There should be 9 questions in the term end examinations question paper.
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.

Course Objectives :

1. To know the evolution, principles and architecture of cellular systems and various concepts associated with mobility management and network signaling
2. To study design and operation of various 2G systems (GSM/GPRS/IS-95/UMTS).
3. To know features and technical aspects of 3G/4G standards evolved from IMT 2000 vision.
4. To know technical standards related to 4G, 5G, Bluetooth, Zigbee, NFC, WiMAX.

Course Outcomes (CO)

- | | |
|-------------|---|
| CO 1 | Understand the evolution, principles and architecture of cellular systems and various concepts associated with mobility management and network signaling. |
| CO 2 | Develop an understanding of design and operation of various 2G systems (GSM/GPRS/IS-95/CDMA). |
| CO 3 | Analyze the features and technical aspects of 3G standards evolved from IMT 2000 vision |
| CO 4 | Analyze the features and technical aspects of 4G, 5G, Bluetooth, Zigbee, NFC and WiMax standards |

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO 1	3	1	2	3	3	2	2	-	2	2	3	3
CO 2	3	1	2	2	2	2	3	-	3	2	2	3
CO 3	3	3	3	2	3	2	3	-	2	2	3	2
CO 4	3	3	3	3	3	3	2	-	2	3	3	2

UNIT I

Introduction To Wireless Communication Systems: Evolution of mobile radio communications; Overview of generations of cellular systems, comparison of various wireless systems.

Introduction to Personal Communication Services (PCS): PCS architecture, Mobility management, Networks signaling. A basic cellular system, multiple access techniques: FDMA, TDMA, CDMA.

Introduction to Wireless Channels and Diversity: Fast Fading Wireless Channel Modeling, Rayleigh/Ricean, Fading Channels, BER Performance in Fading Channels, Introduction to different modulation technique.

UNIT II

2G Networks: AMPS, ETACS, GSM: GSM Architecture, Mobility Management, Network signalling, mobile management, voice signal processing and coding, 2.5G Mobile Data Networks: Introduction to Mobile Data Networks, General Packet Radio Services (GPRS): GPRS architecture, GPRS Network nodes, EDGE, Mobile IP, CDMA IS-95: Spread Spectrum, Frequency and Channel Specifications, Forward and Reverse CDMA Channel, Near-Far Problem, Power Control, Spread Spectrum Systems Cellular Code Division Access Systems-Principle, Power Control, effects of multipath propagation on code division multiple access.

UNIT III

Third Generation (3G) Mobile Services: The Universal Mobile Telecommunication System (UMTS) & CDMA 2000 standards, UMTS Network Architecture Release 99, UMTS Interfaces, UMTS Network Evolution, UMTS Channels, UMTS Time Slots.

UNIT IV

Evolution of Generations: Features of 4G (LTE & VoLTE), 5G, Bluetooth: Architecture, Feature, Frequency Band, Master-Slave, IEEE Standard, Zigbee: Architecture, Feature, Frequency Band, IEEE Standard, NFC: IEEE Standard, Architecture, Feature, IEEE 802.16 (WiMax): Standard Architecture, Spectrum Allocation, Overview of WiMAX PHY, MAC Layer, Scheduling services, UGS, rtPS, nrtPS, Best Effort (BE).

Textbook(s):

1. Raj Pandya, "Mobile & Personnel communication Systems and Services", Prentice Hall India, 2001.
2. Theodore S. Rappaport, "Wireless Communication- Principles and practices," 2nd Ed., Pearson Education Pvt. Ltd, 5th Edition, 2008.
3. Wireless Networks: Applications and Protocols, T.S Rappaport, Pearson Education.
4. Wireless Communication and Networks: 3G and Beyond, I. Saha Mishra, TMH Education

References:

1. T.L.Singhal "Wireless Communication", Tata McGraw Hill Publication.
2. Jochen Schiller, "Mobile communications," Pearson Education Pvt. Ltd., 2002.
3. Yi –Bing Lin & Imrich Chlamatac, "Wireless and Mobile Networks Architecture," John Wiley & Sons, 2001.
4. Lee, W.C.Y., "Mobile Cellular Telecommunication", 2nd Edition, McGraw Hill, 1998.
5. Smith & Collins, "3G Wireless Networks," TMH, 2007
6. Schiller, Jochen, "Mobile Communications", 2nd Edition, Addison Wesley.

Wireless Communication and Networks Lab			
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1. An introduction to basic digital baseband communication through MATLAB® simulations
2. Simulate Bluetooth voice transmission to observe effect of AWGN and interference of 802.11b on transmission using MATLAB and Simulink.
3. Develop a mobile application for wireless technology using any wizards such as available on www.appypie.com or any other.
4. Simulate the line coding techniques using MATLAB and Simulink
5. Simulate the Binary amplitude shift keying using MATLAB and Simulink
6. Simulate the Binary phase shift keying using MATLAB and Simulink
7. Simulate the Quadrature phase shift keying using MATLAB and Simulink
8. Simulate the Delta Modulation using MATLAB and simulink.
9. Simulate the Adaptive delta Modulation using MATLAB and simulink.
10. Simulate the Direct sequence spread spectrum using MATLAB and Simulink
11. Simulate WSN node to determine position on node and blink LED using cupcarbon simulator and senscript.